

Rings and Modules

Prelude

1.

Find an example of each of the following.

- (a) A ring R and a left ideal $I \subset R$ that is *not* a right ideal.
- (b) A ring R and a prime ideal $P \subset R$ that is not maximal. (Let's make this harder and say $P \neq \{0\}$.)
- (c) A ring R , an ideal I , and a prime ideal $P \subset R/I$.
- (d) A ring R and an ideal P that is primary, but not prime.
- (e) A map of rings $f: R \rightarrow S$ and a prime ideal $P \subset R$ such that $f(P)S \subset S$ is *not* prime. (Can you find an example where $f(P)S \neq (1)$?)

* 2.

Let R be a commutative ring, I an ideal in R , and S a multiplicative subset of R . There are two standard correspondences involving prime ideals:

- (i) Prime ideals in R/I are in bijection with prime ideals in R which contain I .
- (ii) Prime ideals in $S^{-1}R$ are in bijection with prime ideals in R which do not meet S .

Prove (i) and (ii).

* 3.

- (a) Give an example of a ring with finitely many elements that has exactly three prime ideals.
- (b) Give an example of a subring of $\mathbb{Q}$ that has exactly three prime ideals.

3.5

Let R be a local ring with maximal ideal $\mathfrak{m}$. Show that $x \in R$ is a unit if and only if $x \notin \mathfrak{m}$.

Fugue

* 4

This problem involves ideals in a commutative ring (with unit). A proper ideal I is *primary* if whenever $ab \in I$ and $a \notin I$, then $b^n \in I$ for some $n > 0$. The radical of an ideal I , denoted $\sqrt{I}$, is the set $\sqrt{I} = \{f : f^n \in I \text{ for some } n > 0\}$.

- (a) Prove that if I is primary, then $\sqrt{I}$ is prime.
- (b) Is the ideal $J = (x^2, xy) \subset \mathbb{Q}[x, y]$ primary? Be sure to carefully justify your answer.

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Let R be a commutative ring with unity. Show that a polynomial

$$f(t) = c_n t^n + c_{n-1} t^{n-1} + \cdots + c_0 \in R[t]$$

is nilpotent if and only if all of its coefficients $c_0, \dots, c_n \in R$ are nilpotent.

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Let R be a commutative ring, and S a multiplicative subset of R .

- (a) When is $S^{-1}R = 0$?
- (b) When is the localization map $R \rightarrow S^{-1}R$ injective?

Quick notational aside: remember that for a commutative ring R and an element $f \in R$, we write R_f for the localization of R at the multiplicative set $\{f^n : n \geq 0\}$. HOWEVER! This is confusing because for a prime ideal $P \subset R$, we write R_P for the localization of R at the multiplicative set $R \setminus P$. So $\mathbb{C}[x]_x$ is different from $\mathbb{C}[x]_{(x)}$. Worse yet, people often write $\mathbb{Z}_p$ to mean either the p -adic integers, or $\mathbb{Z}/p\mathbb{Z}$.

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- (a) Give an example of a commutative ring R and a nonzero element $f \in R$ where the localization $R_f = 0$.
- (b) Give an example of a commutative ring R and a nonzero element $f \in R$ where the localization map $R \rightarrow R_f$ is neither injective nor surjective.
- (c) Give an example of a *local* ring R and an element $f \in R$ where $R_f \neq 0$ but R_f is no longer a local ring.

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Consider $M_2(\mathbb{Z})$, the ring of 2×2 matrices with entries in $\mathbb{Z}$.

- (a) * Find a 2-sided ideal of $M_2(\mathbb{Z})$ that is neither 0 nor the whole ring.
- (b) * Give an example of a nonzero left zero-divisor of $M_2(\mathbb{Z})$.
- (c) * Show that the left zero-divisors of $M_2(\mathbb{Z})$ do not form an ideal.
- (d) Find all the prime ideals of $M_2(\mathbb{Z})$. (For a noncommutative ring, a 2-sided ideal P is prime if whenever A, B are 2-sided ideals such that $AB \subset P$, then either $A \subset P$ or $B \subset P$.)

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Consider the ring $R = \mathbb{C}[x]/(x^3)$.

- (a) Give an example of a nonzero zero-divisor in R .
- (b) Show that the set of zero-divisors in R form an ideal.
- (c) Let S be a commutative ring. Prove that if the set of zero-divisors is an ideal I , then $I \subset S$ is a prime ideal.